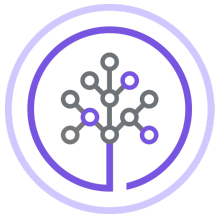


Lab: Exploring Networking Ports and Protocols



Skills
Network

Estimated time needed: **20** minutes

In this lab, you will investigate local and online resources to see how they use protocols, and to find public and private IP addresses.

Objectives

In this hands-on lab, you will:

- Identify the protocols used by the default web browser
- Identify the protocol used when visiting a website
- Determine the computer's public IP address
- Determine the computer's private IP address

Important Notices about This Lab

About Lab Sessions

Lab sessions are not persisted. This means that every time you connect to this lab, a new environment is created for you. Any data or files you saved in a previous session are no longer available. To avoid losing your data, plan to complete these tasks in a single session.

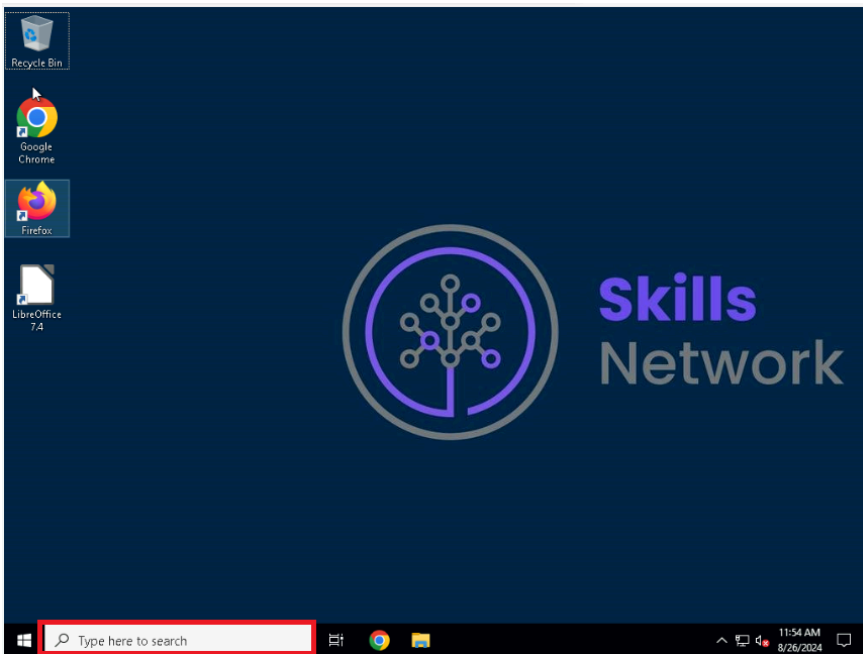
About the Lab Instructions and Solutions

In case you try to use your physical keyboard in the lab environment, it might not produce any visible results. To avoid this issue, please use the **On-Screen Keyboard** (you can find it by searching for **On-Screen Keyboard** in the search bar at the bottom of your screen). If search functionality doesn't work, you can also click on the **Windows** icon, scroll down to find **Windows Ease of Access**, click on it, and then select **On-Screen Keyboard**.

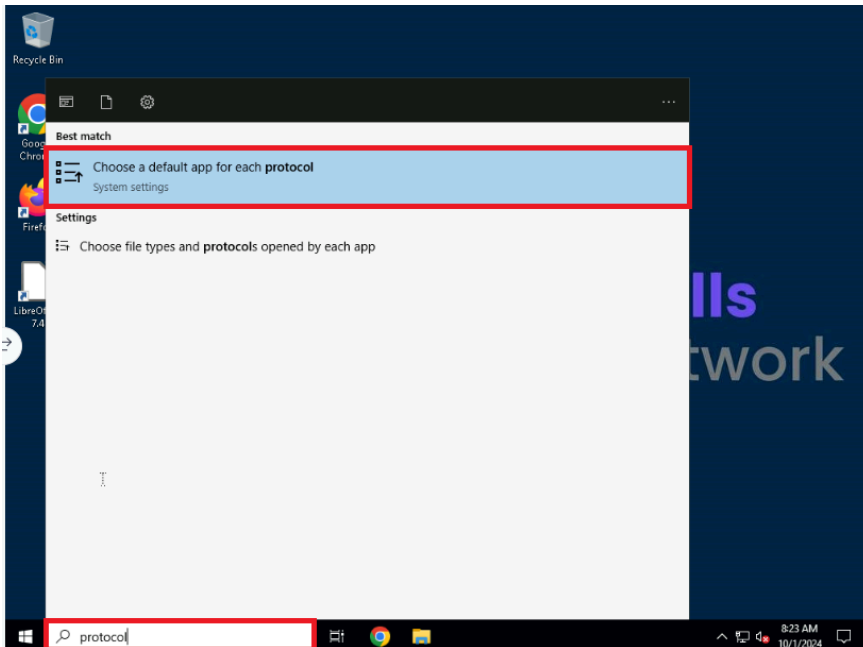
Microsoft Windows operating system features can vary based on the Windows edition. If completing these exercises on your machine, your navigation and solutions may differ from what's presented in this lab.

Exercise 1: Identify the Default Applications Used for The HTTP and HTTPS Protocols

1. Select **Search**.

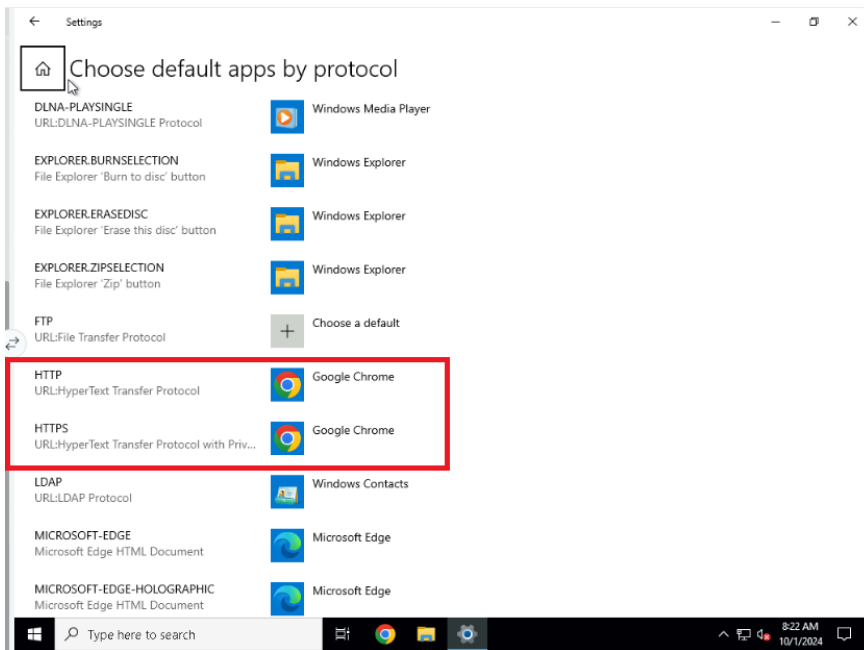


2. Type the word **protocol**, then, select **Choose a default app for each protocol**.



3. Locate the protocols **HTTP** and **HTTPS** and identify the default application.

- The default application for **HTTP** and **HTTPS** is **Google Chrome**.



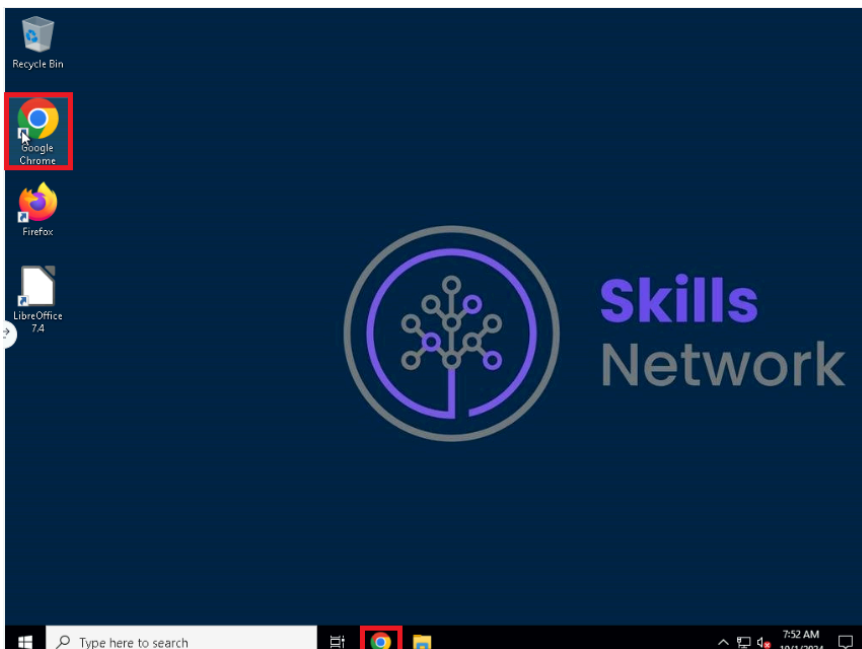
Practice Exercise

Problem: Determine the default application for LDAP.

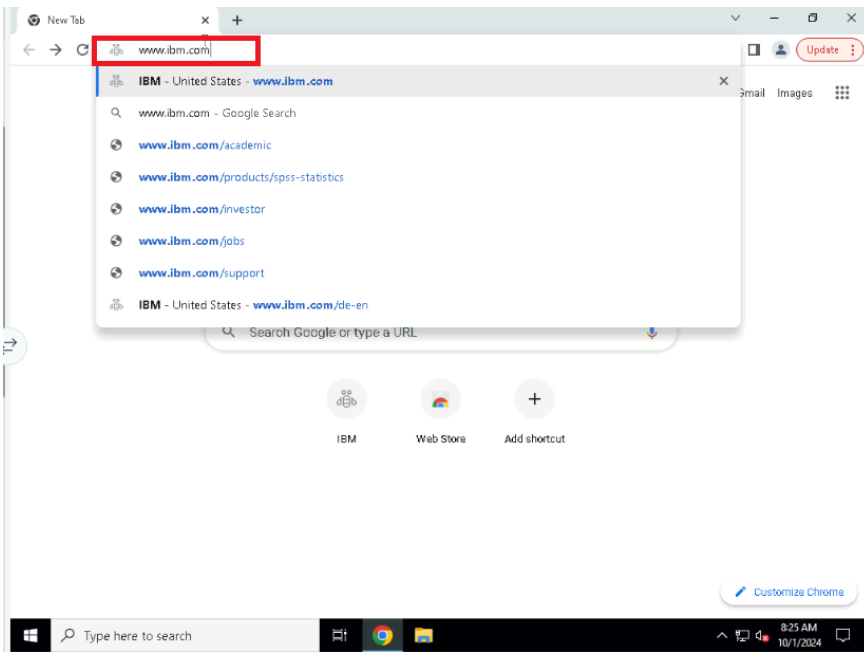
► Click here for the solution.

Exercise 2: Identify the Protocol Used When Visiting a Website

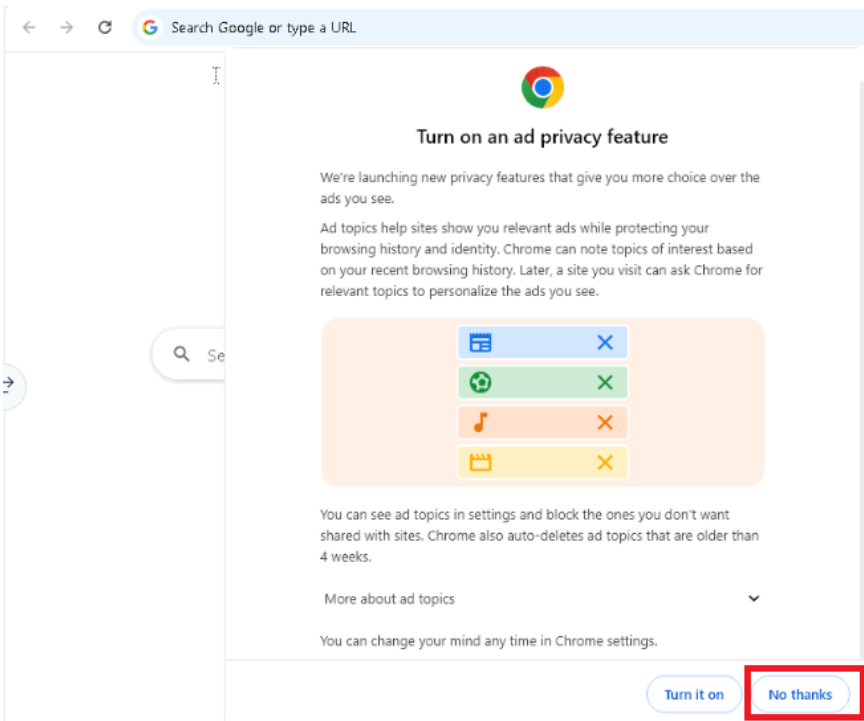
1. Locate the **Google Chrome** application and **double-click** the application to open.



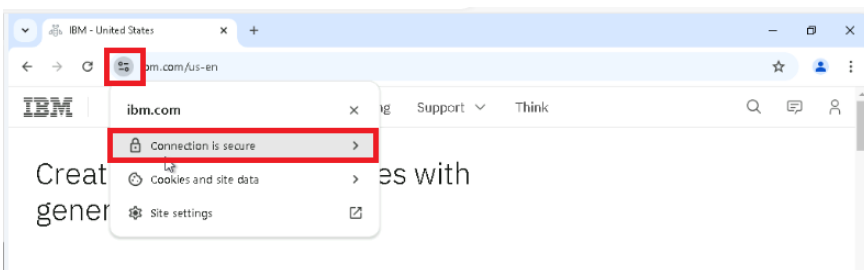
2. Within the search bar, type www.ibm.com, then press **Enter**.



***Note:** If you get any notification while opening the website please scroll down and chose "No Thanks" and proceed.*

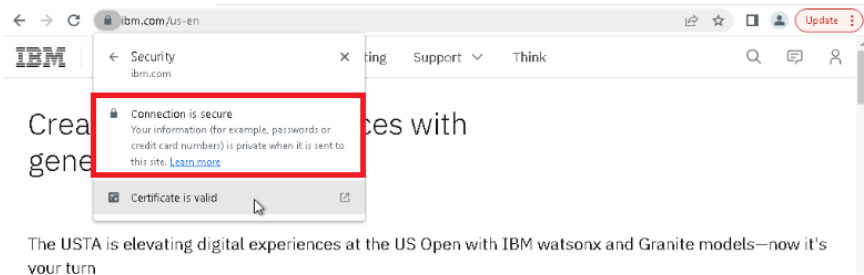


3. Once the website loads, click the **Site Information** icon. You might also get a **Lock** icon instead of **Site Information** icon here.



4. Notice that the dropdown shows that the connection is secure. This means that the Google Chrome browser is using the HTTPS protocol on port 443 to connect to www.ibm.com. We know this information because HTTPS uses additional security measures to protect the data transmitted on the webpage and because port 443 is used solely for secure traffic.

Note: Websites that use the HTTP protocol on port 80 are insecure. Insecure connections are far more vulnerable to hackers and malware than secure connections.



Practice Exercise

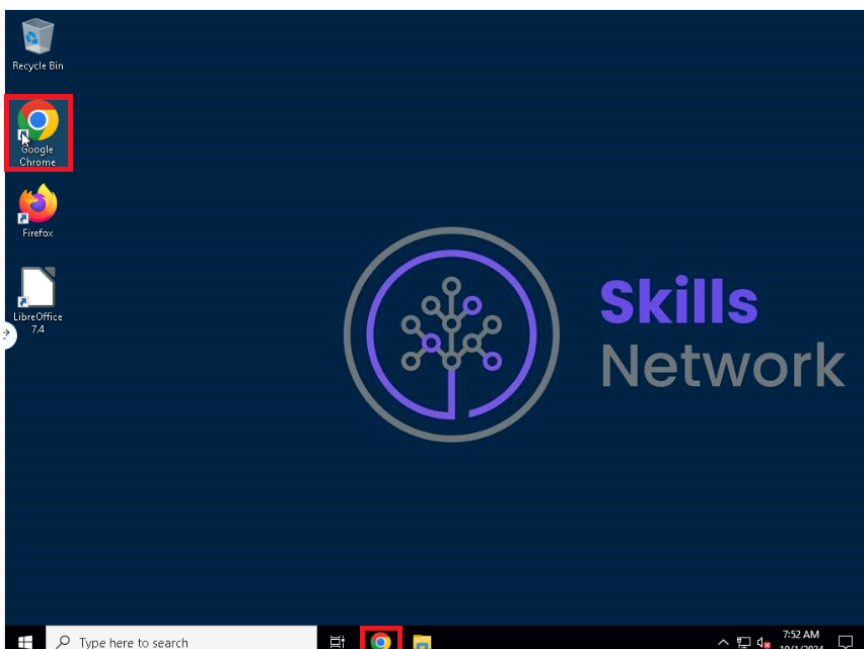
Problem: Determine if the connection to www.lenovo.com is secure.

► Click here for the solution.

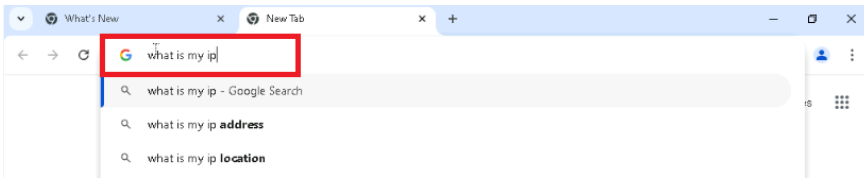
Exercise 3: Determine the Computer's Public IP Address

A public IP address is the address used to access the internet. Public IP addresses are provided by an Internet Service Provider (ISP).

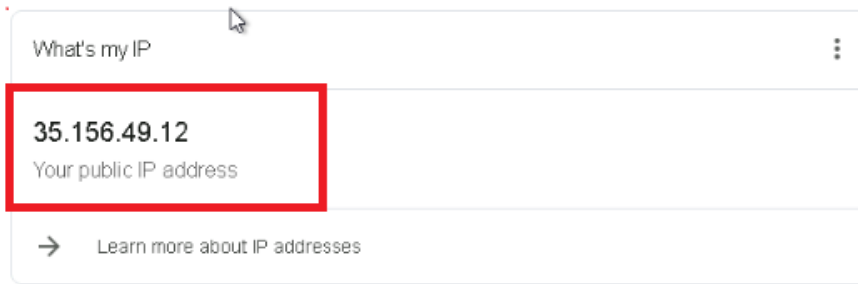
1. Locate the **Google Chrome** application and double-click the application to open.



2. Within the search bar, type **What is my ip**, then press **Enter**.



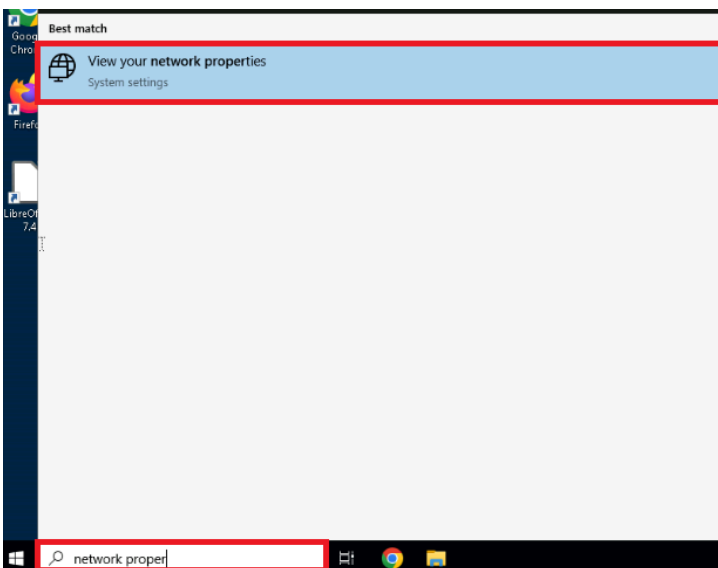
3. Scroll down to see Chrome displays the computer's public IP address.



Exercise 4: Determine the Computer's Private IP Address

A private IP address is the address used to access LAN resources. A device's private IP address is only used within a local network.

1. Search for **Network Properties** and select **View your network properties**



*Note: You can also go to **Settings**, click **Network & Internet**, and then select **Status->Advanced Network Settings**. Then select **View hardware and Connection Properties***

2. Review the left side of the screen.

3. Locate **IPv4 address** and then **IPv6 address**.

[← Settings](#)

🏠 View hardware and connection properties

Link speed (Receive/Transmit):	25/25 (Gbps)
DHCP enabled:	Yes
DHCP servers:	172.31.48.1
DHCP lease obtained:	Tuesday, August 27, 2024 7:46:04 AM
DHCP lease expires:	Tuesday, August 27, 2024 8:46:04 AM
IPv4 address:	172.31.62.131/20
IPv6 address:	fe80::815d:76b0:8e36:2142%7/64
IPv4 default gateway:	172.31.48.1
IPv6 default gateway:	
DNS servers:	172.31.0.2 (Unencrypted)
DNS domain name:	eu-central-1.compute.internal
DNS connection suffix:	eu-central-1.compute.internal
DNS search suffix list:	
Network name:	Network 3
Network category:	Public
Connectivity (IPv4/IPv6):	Connected to Internet / Connected to unknown network

4. You have found the computer's private IPv4 and IPv6 IP addresses.

Congratulations! You have completed this lab and are ready for the next topic.

Authors

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